

Replicator dynamics

1. The replicator equation for the proportion x_i of type i is ...
 - (a) $\dot{x}_i = \phi - f_i(x)$
 - (b) $\dot{x}_i = x_i \phi$
 - (c) $\dot{x}_i = x_i(f_i(x) - \phi)$, where ϕ is the average fitness
 - (d) $\dot{x}_i = f_i(x)$
2. A type's frequency increases under the replicator dynamics when ...
 - (a) it is rare
 - (b) the average fitness is zero
 - (c) its fitness is below the population average
 - (d) its fitness is above the population average
3. An evolutionarily stable strategy (ESS) is one that ...
 - (a) every individual must play
 - (b) maximises the total payoff of the population
 - (c) cannot be invaded by a small group playing a different strategy
 - (d) is always a pure strategy
4. A rest point (stable population) of the replicator dynamics satisfies ...
 - (a) $f_i(x) = 0$ for all i
 - (b) $\phi = 1$
 - (c) $\dot{x}_i = 0$ for all i
 - (d) $x_i = 1$ for some i
5. Every ESS is also ...
 - (a) a zero sum outcome
 - (b) a strictly dominant strategy
 - (c) a symmetric Nash equilibrium
 - (d) an unstable rest point
6. In the Hawk-Dove game with $V < C$, the dynamics approach ...
 - (a) all Doves
 - (b) no rest point
 - (c) an interior mix of Hawks and Doves (the ESS)
 - (d) all Hawks

7. The average population fitness ϕ is ...
 - (a) always equal to 1
 - (b) $\sum_i x_i f_i(x)$, the fitness-weighted average over types
 - (c) the fitness of the fittest type
 - (d) the number of types
8. Under the replicator dynamics, a rare but fit type grows ...
 - (a) only if it is the only type
 - (b) at the same rate as a common type
 - (c) instantly to fixation
 - (d) slowly, because growth is proportional to its current frequency x_i
9. In a pairwise interaction game with payoff matrix M , the fitness of type i is ...
 - (a) $\sum_j M_{ij}$
 - (b) the largest entry of M
 - (c) $f_i(x) = \sum_j x_j M_{ij}$
 - (d) M_{ii}
10. The post-entry population after a fraction ϵ of mutants y invade residents x is ...
 - (a) $x + y$
 - (b) ϵy
 - (c) $x - y$
 - (d) $x + \epsilon(y - x)$
11. In Rock-Paper-Scissors, the replicator dynamics ...
 - (a) converge to the uniform mix
 - (b) converge to all Rock
 - (c) have no rest point
 - (d) cycle around the interior equilibrium rather than converging to it
12. A two-strategy coordination game (such as which side of the road to drive on) is ...
 - (a) without any stable population
 - (b) has two stable populations, each pure, with an unstable interior population between them
 - (c) a zero sum game
 - (d) always convergent to a single mix
13. An ESS is a refinement of Nash equilibrium because ...

- (a) every ESS is a symmetric Nash equilibrium, but not every Nash equilibrium is an ESS
 - (b) they are unrelated
 - (c) every Nash equilibrium is an ESS
 - (d) an ESS need not be a Nash equilibrium
14. The replicator dynamics keep the proportions x_i ...
- (a) constant over time
 - (b) summing to 1 (they remain a valid distribution)
 - (c) summing to the number of types
 - (d) all equal
15. A stable population that the dynamics actually move towards is ...
- (a) the point of maximum fitness
 - (b) any rest point, even a repelling one
 - (c) an asymptotically stable rest point (a small perturbation returns)
 - (d) always $x = 1/2$
16. For a strategy σ^* with $f(\sigma^*, \sigma^*) = f(\sigma, \sigma^*)$, the ESS condition additionally requires ...
- (a) $f(\sigma^*, \sigma) > f(\sigma, \sigma)$
 - (b) $f(\sigma^*, \sigma^*) > f(\sigma, \sigma^*)$
 - (c) $f(\sigma^*, \sigma) < f(\sigma, \sigma)$
 - (d) nothing further
17. Taking $\epsilon \rightarrow 0$ in the ESS definition recovers ...
- (a) the fixation probability
 - (b) a Condorcet cycle
 - (c) the replicator equation
 - (d) the symmetric Nash condition $f(\sigma^*, \sigma^*) \geq f(\sigma, \sigma^*)$
18. The replicator-mutator dynamics extend the replicator equation by adding ...
- (a) pairwise proposals
 - (b) a fixed population size
 - (c) discounting of future payoffs
 - (d) mutation between types
19. The asymmetric replicator dynamics model ...
- (a) two interacting populations with distinct roles and payoff matrices

- (b) a finite population of fixed size
 - (c) a zero-sum game only
 - (d) a single population playing itself
20. In a pairwise interaction game with payoff matrix M , the fitness of playing σ against population state x is ...
- (a) $\sigma + x$
 - (b) $\det(M)$
 - (c) the largest entry of M
 - (d) σMx (linear in the population state)